

Financial Mathematics

(MATH35400/MATHM5400)

Problem Class 8 – 13 March 2026 ¹

1. (2021 exam, level 3, Q3 (b-c))

(b) Let $(W_t)_{t \geq 0}$ be standard Brownian motion.

- i. Calculate the probability that $W_{10} \geq 5$.
- ii. Show that for any $s, t \geq 0$ the covariance satisfies $\text{Cov}(W_s, W_t) = \min(s, t)$.
- iii. Define a new process by $B_0 = 0$ and $B_t = tW_{1/t}$ for $t > 0$. Show that $\text{Cov}(B_s, B_t) = \min(s, t)$.

(c) Consider a stock process of the form

$$S_t = \exp(\sigma W_t + (\mu - \sigma^2/2)t),$$

and a bond process $B_t = \exp(rt)$. For any $t \geq 0$, find the probability that the stock is more valuable than the bond at time t . Find the limit of this probability as $t \rightarrow \infty$.

2. (2020 exam, Q3 (c))

- (i) Find the probability that standard Brownian motion lies between some values a and b (where $a \leq b$) at time T , in terms of the normal distribution function Φ . Find the expected value of $\exp(cW_T)$ for any c .

[Hint: You may assume that if $Z \sim \mathcal{N}(0, \sigma^2)$, then it has moment generating function $M_Z(t) = \mathbb{E}(\exp(tZ)) = \exp(t^2\sigma^2/2)$.]

- (ii) Consider two independent standard Brownian motions $W_t^{(1)}$ and $W_t^{(2)}$, and write the linear combination

$$X_t = \gamma W_t^{(1)} + \sqrt{1 - \gamma^2} W_t^{(2)},$$

where $-1 \leq \gamma \leq 1$ is a constant. Show that X_t is itself a standard Brownian motion.

3. (Problem Sheet 7, 2021/22)

Consider a Cox-Ross-Rubinstein model with $T = 3$ periods, $S_0 = 100$, $u = 1.6$ and $d = 0.6$. The interest rate is $r = 0.1$. What is the time 0 price of an American put that has exercise price $e = 90$? Show that it is only optimal to exercise early in the case that the stock falls twice in the first two moves.

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